

Practical - 2

Practical Title: Semantic HTML5 Pages with Accessibility-Ready Structure

Problem Definition : Develop static HTML5 skeletons for at least 10 StudentHub pages such as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. Use semantic tags and accessibility-friendly structure.

A. Theory

1. Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?

Yes. Semantic HTML5 tags are used according to their purpose. <header> contains the website heading and navigation, <nav> contains page links, <main> contains the main page content, <section> groups related content, <article> contains individual content blocks, <aside> contains additional information, and <footer> contains copyright information.

Source: https://www.w3schools.com/html/html5_semantic_elements.asp

2. Are form labels, headings, image alt text, and navigation links accessible?

Yes. Forms use <label> elements connected to their input fields using the for and id attributes. Each page has a clear <h1> and uses <h2> and <h3> for subheadings. Navigation is provided using normal <a> links inside <nav>. If images are added, meaningful alt text will be used. The lang="en" attribute and aria-label for navigation also improve accessibility.

3. Is the page structure consistent across all pages?

Yes, the page structure is consistent across all 10 pages.

All 10 StudentHub pages follow the same basic structure:

Header → Navigation → Main Content → Aside → Footer.

This makes the website easy to navigate, understand, and maintain.

The navigation menu is kept similar on all pages and makes the website easy to use and maintain.

B. Implementation/ Output/Result:

The following are all the tags which I used in the webpages:

• Tag	• Description / How it was used
• <meta>	• Provides metadata like character encoding and responsive viewport settings.
• <title>	• Sets the title displayed on the browser tab.
• <header>	• Used at the top of every page for the StudentHub title and navigation.
• <nav>	• Contains links to move between the 10 StudentHub pages.
• <main>	• Contains the main content specific to each page.
• <section>	• Divides the main content into logical sections such as Features, Results, or Profile.
• <article>	• Used for individual content blocks such as courses, assignments, and academic information.
• <aside>	• Used for additional information such as notices, reminders, and quick information.
• <footer>	• Used at the bottom of pages for copyright information.
• <h1>	• Used for the main heading, usually "StudentHub Portal".
• <h2>	• Used for major section headings such as "My Courses" or "Student Login".
• <h3>	• Used for smaller headings such as individual courses or features.
• <p>	• Used to display paragraphs and simple student information.
• <a>	• Used for navigation between different HTML pages and other links.
• <form>	• Used to create Login, Registration, and Contact forms.
• <label>	• Provides a name for form fields and improves accessibility.
• <input>	• Used for entering details such as name, email, password, ID, phone, etc.
• <select>	• Used in the Register page to select the student's program.

- | • Tag | • Description / How it was used |
|---------------------------------|--|
| • <code><option></code> | • Provides choices inside the <code><select></code> element, such as B.Tech, BCA, and MCA. |
| • <code><textarea></code> | • Used in the Contact page for entering a longer message. |
| • <code><button></code> | • Used for actions such as Login, Create Account, and Send Message. |
| • <code><table></code> | • Used to display attendance and semester result information in rows and columns. |
| • <code><caption></code> | • Provides a title/description for the Attendance and Results tables. |
| • <code><thead></code> | • Contains the heading row of the tables. |
| • <code><tbody></code> | • Contains the actual attendance or result data. |
| • <code><tr></code> | • Represents a row in a table. |
| • <code><th></code> | • Represents table headings such as Subject, Present, Total, and Grade. |
| • <code><td></code> | • Contains individual table data such as marks and attendance percentages. |
| • <code></code> | • Can be used to create an unordered list, although the current simple navigation mainly uses links. |
| • <code></code> | • Represents an individual item inside a list. |
| • <code>
</code> | • Used in the Contact form to place labels and fields on separate lines. |

C. Evidence of Work (Screenshots):

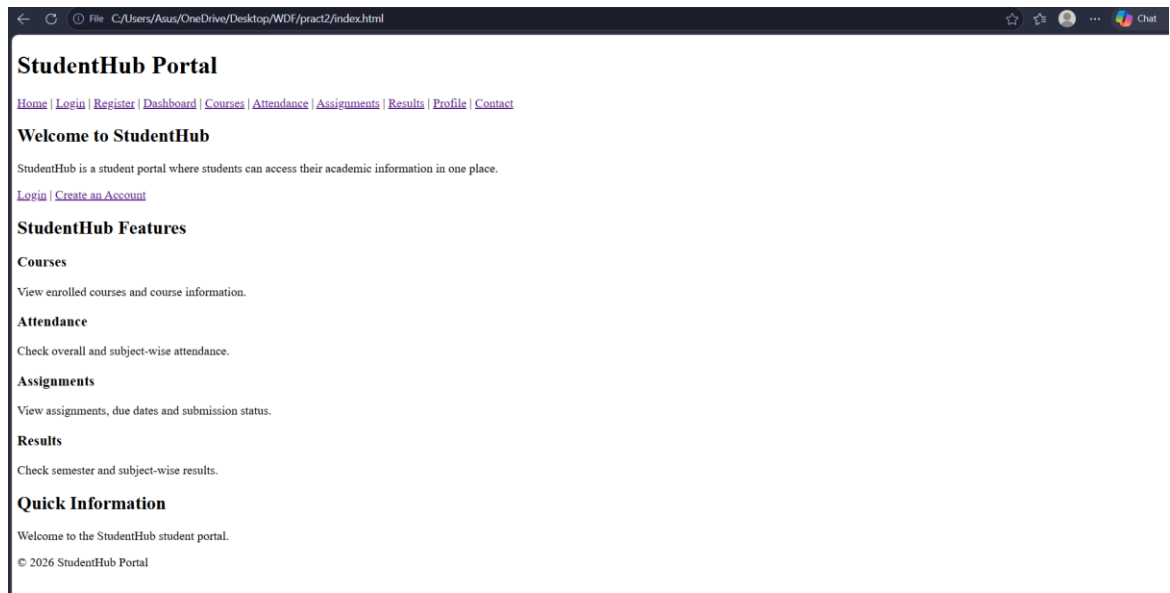


Fig1: Index

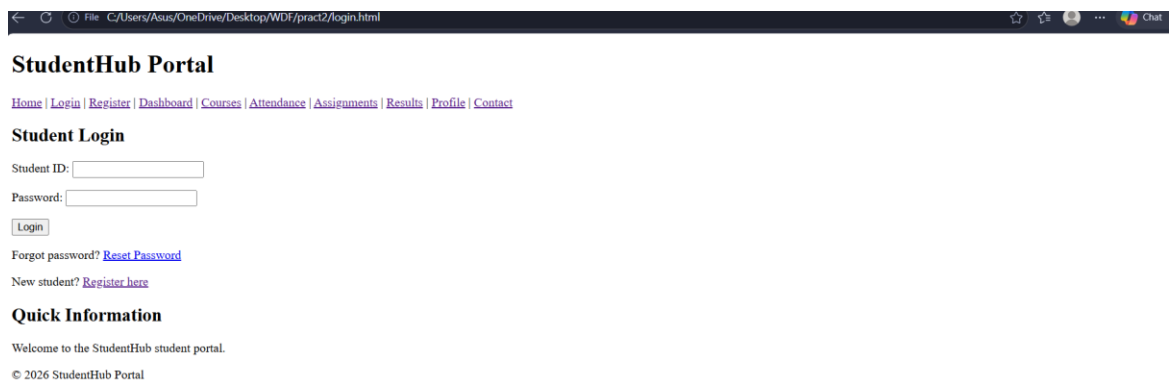
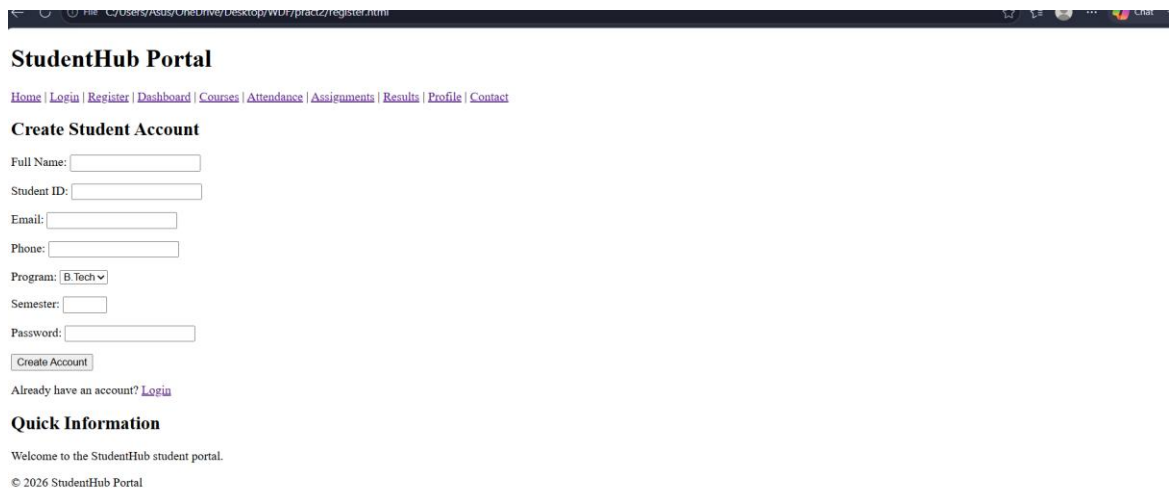


Fig2: Login



The screenshot shows the 'StudentHub Portal' registration page. At the top, there is a navigation bar with links: Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, and Contact. Below the navigation bar, the main heading is 'Create Student Account'. This section contains several input fields: 'Full Name', 'Student ID', 'Email', 'Phone', 'Program' (a dropdown menu currently showing 'B.Tech'), 'Semester', and 'Password'. A 'Create Account' button is located below these fields. Below the button, there is a link 'Already have an account? Login'. At the bottom of the page, there is a 'Quick Information' section with the text 'Welcome to the StudentHub student portal.' and a copyright notice '© 2026 StudentHub Portal'.

StudentHub Portal

[Home](#) | [Login](#) | [Register](#) | [Dashboard](#) | [Courses](#) | [Attendance](#) | [Assignments](#) | [Results](#) | [Profile](#) | [Contact](#)

Create Student Account

Full Name:

Student ID:

Email:

Phone:

Program:

Semester:

Password:

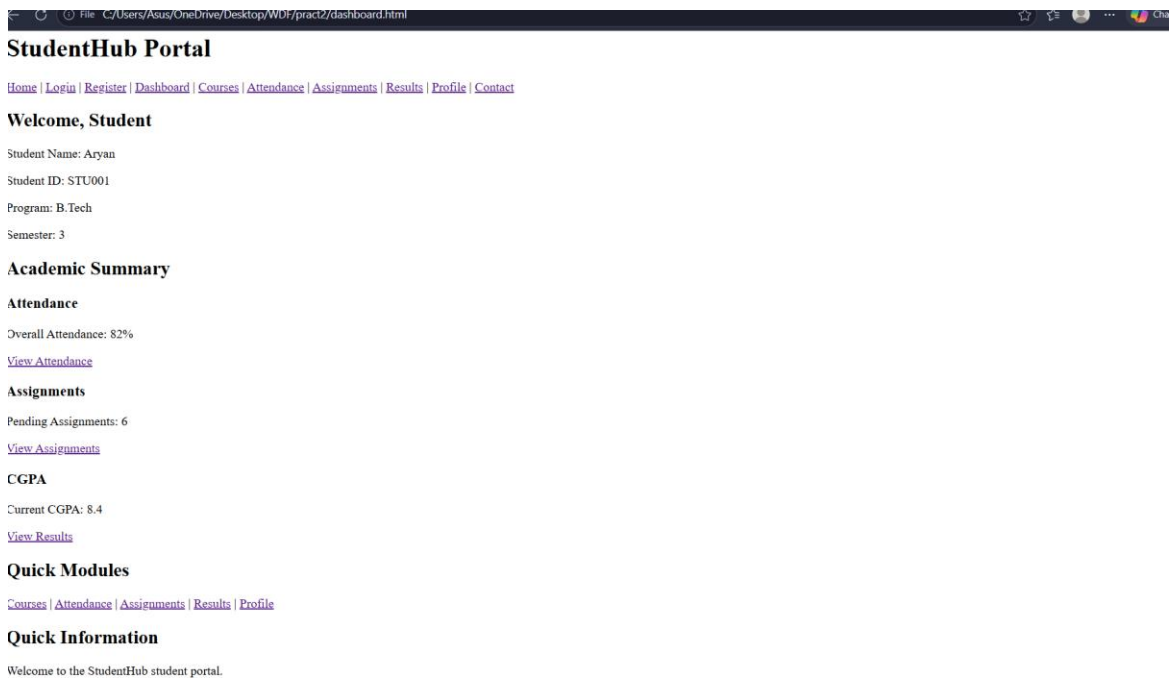
Already have an account? [Login](#)

Quick Information

Welcome to the StudentHub student portal.

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Fig3: Register



The screenshot shows the 'StudentHub Portal' dashboard page. At the top, there is a navigation bar with links: Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, and Contact. Below the navigation bar, the main heading is 'Welcome, Student'. This section displays the student's information: 'Student Name: Aryan', 'Student ID: STU001', 'Program: B.Tech', and 'Semester: 3'. Below this, there is an 'Academic Summary' section with three sub-sections: 'Attendance' (Overall Attendance: 82%, with a 'View Attendance' link), 'Assignments' (Pending Assignments: 6, with a 'View Assignments' link), and 'CGPA' (Current CGPA: 8.4, with a 'View Results' link). Below the 'Academic Summary' section, there is a 'Quick Modules' section with links: Courses, Attendance, Assignments, Results, and Profile. At the bottom of the page, there is a 'Quick Information' section with the text 'Welcome to the StudentHub student portal.'.

StudentHub Portal

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Welcome, Student

Student Name: Aryan

Student ID: STU001

Program: B.Tech

Semester: 3

Academic Summary

Attendance

Overall Attendance: 82%

[View Attendance](#)

Assignments

Pending Assignments: 6

[View Assignments](#)

CGPA

Current CGPA: 8.4

[View Results](#)

Quick Modules

[Courses](#) | [Attendance](#) | [Assignments](#) | [Results](#) | [Profile](#)

Quick Information

Welcome to the StudentHub student portal.

Fig4: Dashboard

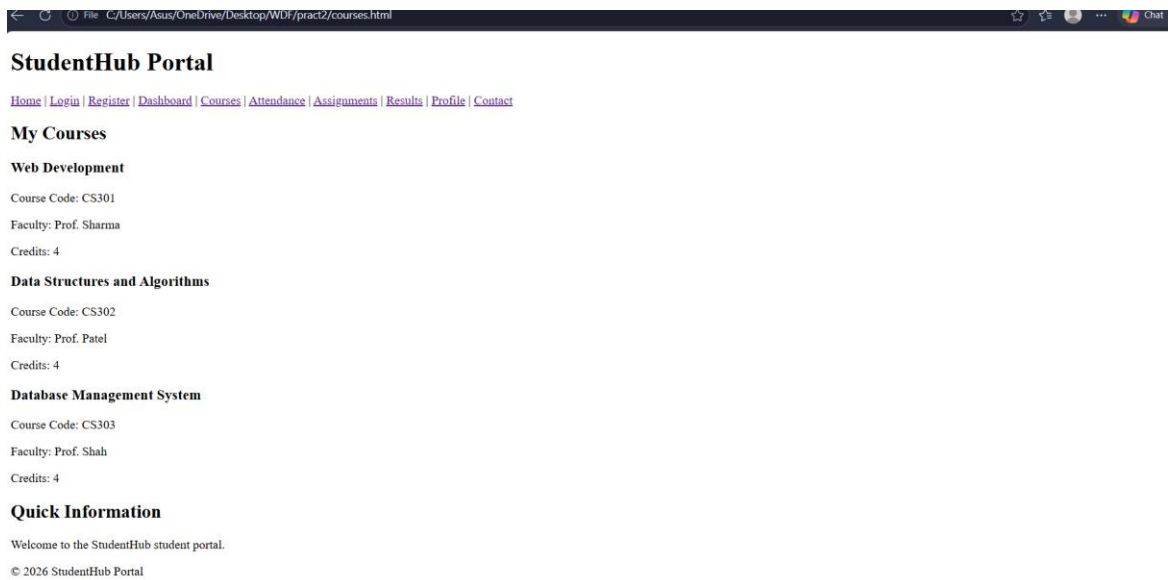


Fig5: Courses

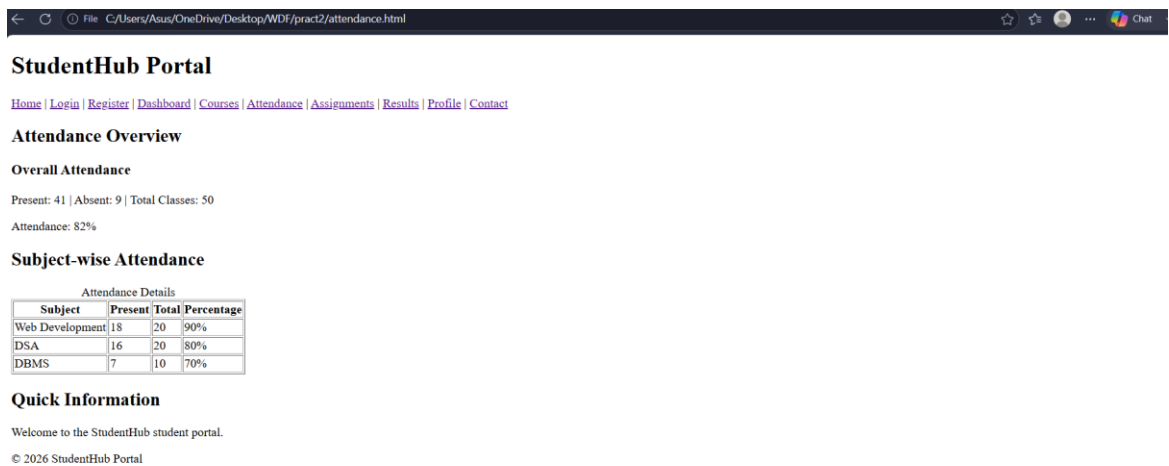
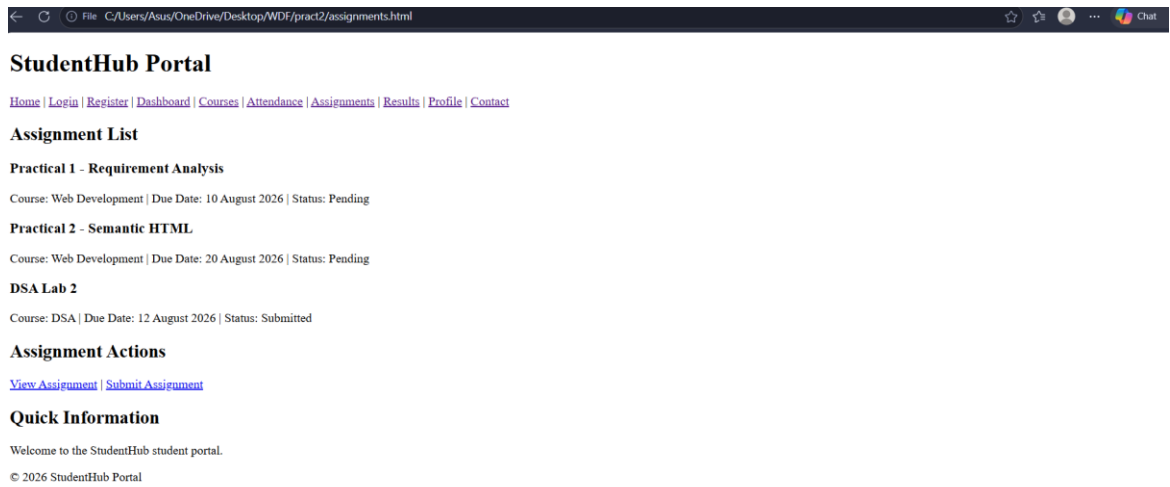


Fig6: Attendance



The screenshot shows a web browser window with the address bar displaying 'C:/Users/Asus/OneDrive/Desktop/WDF/pract2/assignments.html'. The page title is 'StudentHub Portal'. Below the title is a navigation menu with links: Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, and Contact. The main content area is titled 'Assignment List' and contains three sections: 'Practical 1 - Requirement Analysis' with course 'Web Development', due date '10 August 2026', and status 'Pending'; 'Practical 2 - Semantic HTML' with course 'Web Development', due date '20 August 2026', and status 'Pending'; and 'DSA Lab 2' with course 'DSA', due date '12 August 2026', and status 'Submitted'. Below these is an 'Assignment Actions' section with links 'View Assignment' and 'Submit Assignment'. At the bottom is a 'Quick Information' section with a welcome message and a copyright notice for 2026 StudentHub Portal.

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Assignment List

Practical 1 - Requirement Analysis

Course: Web Development | Due Date: 10 August 2026 | Status: Pending

Practical 2 - Semantic HTML

Course: Web Development | Due Date: 20 August 2026 | Status: Pending

DSA Lab 2

Course: DSA | Due Date: 12 August 2026 | Status: Submitted

Assignment Actions

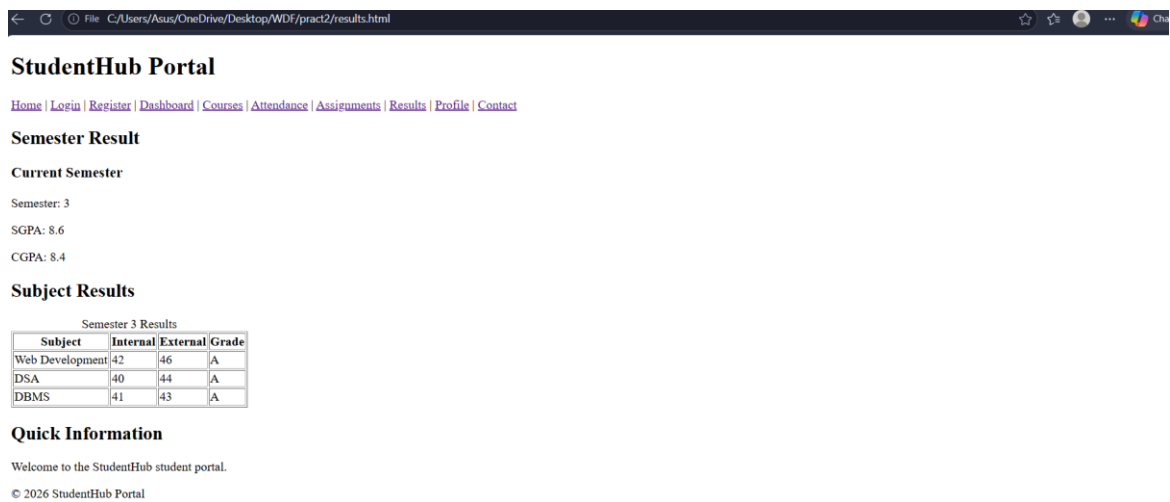
[View Assignment](#) | [Submit Assignment](#)

Quick Information

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Fig7: Assignment



The screenshot shows a web browser window with the address bar displaying 'C:/Users/Asus/OneDrive/Desktop/WDF/pract2/results.html'. The page title is 'StudentHub Portal'. Below the title is a navigation menu with links: Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, and Contact. The main content area is titled 'Semester Result' and contains two sections: 'Current Semester' showing 'Semester: 3', 'SGPA: 8.6', and 'CGPA: 8.4'; and 'Subject Results' showing a table for 'Semester 3 Results'. The table has columns for Subject, Internal, External, and Grade, with rows for Web Development, DSA, and DBMS. Below the table is a 'Quick Information' section with a welcome message and a copyright notice for 2026 StudentHub Portal.

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Semester Result

Current Semester

Semester: 3

SGPA: 8.6

CGPA: 8.4

Subject Results

Semester 3 Results

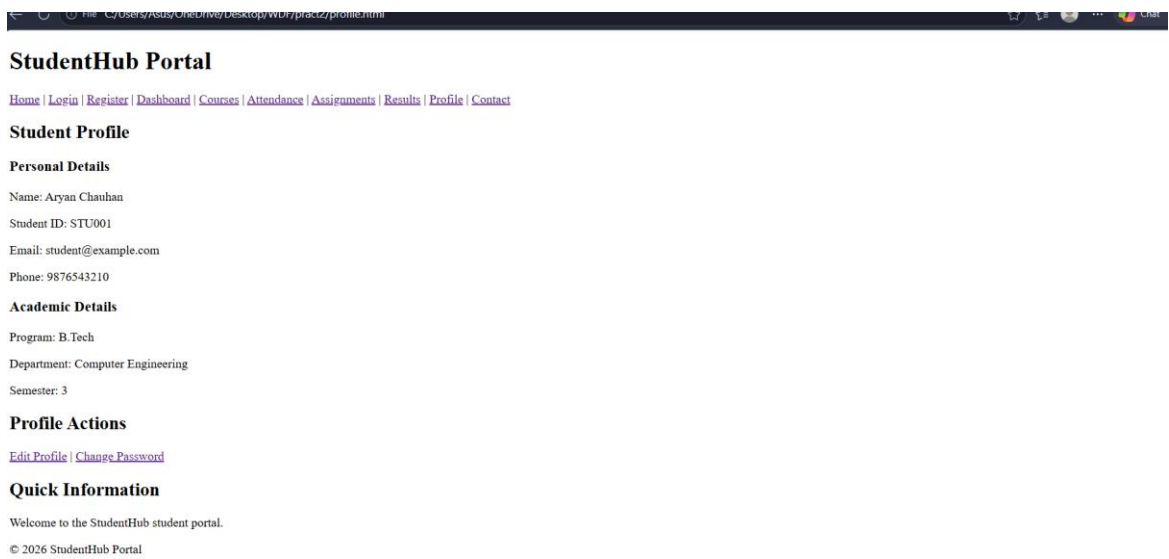
Subject	Internal	External	Grade
Web Development	42	46	A
DSA	40	44	A
DBMS	41	43	A

Quick Information

Welcome to the StudentHub student portal.

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Fig8: Result



The screenshot shows a web browser window with the address bar displaying 'C:/Users/Asus/OneDrive/Desktop/Wd1/pract2/profile.html'. The page title is 'StudentHub Portal'. Below the title is a navigation menu with links: Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, and Contact. The main content area is titled 'Student Profile' and contains three sections: 'Personal Details' with fields for Name (Aryan Chauhan), Student ID (STU001), Email (student@example.com), and Phone (9876543210); 'Academic Details' with fields for Program (B.Tech), Department (Computer Engineering), and Semester (3); and 'Profile Actions' with links for Edit Profile and Change Password. A 'Quick Information' section at the bottom contains a welcome message and a copyright notice for 2026 StudentHub Portal.

StudentHub Portal

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Student Profile

Personal Details

Name: Aryan Chauhan
Student ID: STU001
Email: student@example.com
Phone: 9876543210

Academic Details

Program: B.Tech
Department: Computer Engineering
Semester: 3

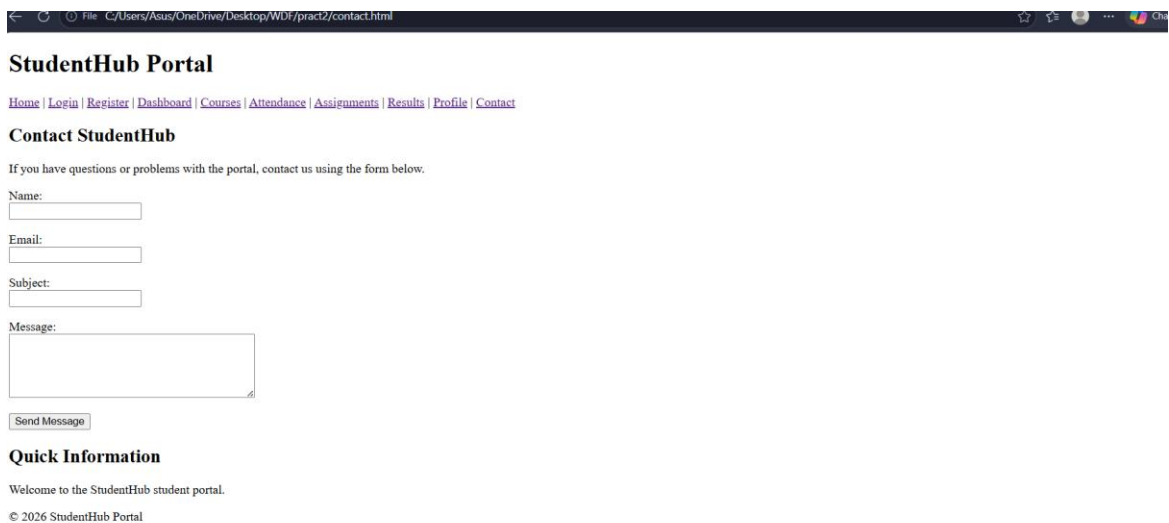
Profile Actions

[Edit Profile](#) | [Change Password](#)

Quick Information

Welcome to the StudentHub student portal.
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Fig9: Profile



The screenshot shows a web browser window with the address bar displaying 'C:/Users/Asus/OneDrive/Desktop/Wd1/pract2/contact.html'. The page title is 'StudentHub Portal'. Below the title is a navigation menu with links: Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, and Contact. The main content area is titled 'Contact StudentHub' and contains a form for contacting the portal. The form includes fields for Name, Email, Subject, and a large text area for the Message. A 'Send Message' button is located below the message field. A 'Quick Information' section at the bottom contains a welcome message and a copyright notice for 2026 StudentHub Portal.

StudentHub Portal

[Home](#) | [Login](#) | [Register](#) | [Dashboard](#) | [Courses](#) | [Attendance](#) | [Assignments](#) | [Results](#) | [Profile](#) | [Contact](#)

Contact StudentHub

If you have questions or problems with the portal, contact us using the form below.

Name:

Email:

Subject:

Message:

Quick Information

Welcome to the StudentHub student portal.
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Fig10: Contact

D. GitHub Repository Link:

[StudentHubPortal/html at main · aryan150806-wq/StudentHubPortal](https://StudentHubPortal/html%20at%20main%20%20aryan150806-wq/StudentHubPortal)

E. Challenges Faced:

Challenges Faced While Creating the Pages

- **Understanding Semantic Tags** – It was initially difficult to understand where to use tags like `<section>`, `<article>`, and `<aside>` correctly.
- **Page Linking** – Connecting all 10 HTML pages using the correct file names and paths was challenging at first.
- **Maintaining Consistent Structure** – Keeping the same header, navigation, and footer across all pages required careful editing.
- **Creating Forms** – Understanding how `<form>`, `<label>`, `<input>`, and `<textarea>` work together took some practice.
- **Creating Tables** – Structuring attendance and result data using `<table>`, `<thead>`, `<tbody>`, `<tr>`, `<th>`, and `<td>` was initially confusing.
- **Accessibility** – Learning to use labels, alt text, aria-label, and proper heading order was a new concept.
- **Managing Multiple Files** – Handling 10 separate HTML files and checking that every link worked correctly was time-consuming.

F. Learning Outcomes:

Through this practical, I learned how to create a basic multi-page website using **HTML5 semantic elements**. I understood the purpose and proper use of tags such as `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, and `<footer>`. I also learned how to create and connect 10 StudentHub pages according to the sitemap and wireframe, while maintaining a consistent page structure and navigation system.

I also gained practical knowledge of creating **forms, tables, headings, links, and input fields** using HTML5. The practical helped me understand basic web accessibility by using proper form labels, heading hierarchy, navigation labels, and required fields. Overall, I developed a foundation for creating **structured, accessible, user-friendly, and maintainable static web pages**.

Faculty Remarks: